Software Engineering & Design Principal Lab

Experiment 2 Report

Experiment: Identification of Problem, Objectives, Scope and Feasibility Analysis

Student/Team Details

Particular	Details
Experiment No.	6
Project Title	Loan Eligibility Prediction System (LEPS)
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Semester & Section	5 - D
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1. Problem Identification

1.1 Background / Context

This project is set in the Banking, Financial Services domain, specifically consumer lending. Loan eligibility assessment sits at the intersection of financial technology and regulatory compliance, since a lender's decision to grant or deny credit directly affects both business risk and a customer's financial well-being. Banks, NBFCs (Non-Banking Financial Companies) and digital lenders increasingly combine manual review, rule-based policy engines and machine-learning models to evaluate applications for products such as personal and consumer loans. This creates an opportunity for a software system that makes the assessment process faster, more consistent, and transparent to both the officer making the decision and the applicant affected by it.

1.2 Existing Problem

Applicants have no reliable way to know whether they are likely to qualify before applying, so many applicants risk a hard credit inquiry for nothing and receive opaque rejections with no actionable reason. Loan officers face a high volume of manual file review and lack tooling to explain why a borderline application was scored the way it was, or to explore “what-if” scenarios during a counselling conversation. At the business level, underwriting cost scales with headcount rather than volume, turnaround time lags behind faster fintech competitors, and inconsistent risk assessment produces both avoidable defaults and avoidable rejections of genuinely good customers. Risk and compliance teams, in turn, struggle to see portfolio-level risk in real time and to reconstruct, months later, exactly why a specific decision was made.

1.3 Existing Method / Current System

The typical evaluation flow today runs through five stages: (1) application intake, where the applicant supplies personal, employment, income and loan-purpose details; (2) a credit-bureau pull, where the lender queries a bureau (e.g., CIBIL, Experian, CRIF High Mark) for a score and report; (3) scorecard/policy evaluation, where the bureau score and application data are checked against hard eligibility rules and a weighted, often hand-built, points-based scorecard; (4) manual review for applications near a policy threshold, of high value, or with flagged inconsistencies; and (5) a decision and disclosure step, where the applicant is informed of an approval, rejection, or counter-offer, generally with only a generic reason given for a decline.

1.4 Limitations of the Existing System

1. Manual and rule-based review is slow, inconsistent across officers and branches, and does not scale with application volume.
2. Traditional scorecards are infrequently recalibrated, coarse-grained, and blind to alternative data, making thin-file applicants hard to assess fairly.
3. Where machine-learning models are already in use, explanations (if offered at all) are often reconstructed after the decision rather than derived from it, so a rejected applicant rarely receives a specific, actionable reason.
4. Audit trails are weak: many systems log only the final outcome, not a reconstructable rationale tied to the exact inputs and model version used at decision time.
5. There is little to no applicant-facing guidance on what would need to change for a declined applicant to become eligible in future.

2. Problem Statement

Loan applicants — particularly thin-file or first-time borrowers — along with loan officers and lenders' risk and compliance functions, currently rely on an assessment process that is either manual and inconsistent, rule/scorecard-based and coarse, or machine-learning-based but opaque. No single existing approach delivers eligibility assessment that is simultaneously accurate, fast, explainable and auditable. As a result, applicants cannot get a trustworthy pre-application read on their chances or guidance on how to improve their eligibility; officers lack tooling to explain or explore borderline decisions; and lenders cannot cheaply scale consistent, defensible decision-making. A software-based solution is required because these limitations stem from process and tooling gaps — volume, consistency, explainability and auditability — that manual effort and legacy scorecards cannot close on their own, and because a growing body of regulatory guidance now expects lenders to produce specific, model-derived reasons for adverse credit decisions rather than generic ones.

3. Objectives of the Project

S. No.	Objective
1.	To develop a system that predicts loan eligibility from applicant, financial and credit-bureau data.
2.	To generate a calibrated eligibility probability and risk tier, rather than a bare yes/no output.
3.	To explain each prediction in plain language, ranking the factors that help and hurt eligibility, using a method faithful to the underlying model.
4.	To provide loan officers a dashboard to review, question and override predictions, with every override logged.
5.	To validate applicant-submitted data for completeness and consistency before a prediction is attempted.
6.	To maintain an immutable, queryable audit trail of every prediction and officer decision.
7.	To identify, for a declined or borderline applicant, the specific factors that would most improve their eligibility if changed.
8.	To monitor model performance and prediction-distribution drift over time.

4. Scope of the Project

4.1 Target Users / Stakeholders

S. No.	User / Stakeholder	Role
1.	Loan Officer / Underwriter	Reviews applications, reviews the system's prediction and explanation, confirms or overrides the decision
2.	Applicant / Borrower	Submits application details, receives the eligibility outcome and, where declined, plain-language guidance
3.	Risk / Credit Policy Team	Sets and calibrates eligibility policy and risk appetite based on portfolio-level outcomes
4.	Compliance / Audit Team	Verifies fair-lending compliance and reconstructs the rationale for any decision on demand
5.	System Administrator	Manages user access, roles, and model version promotion

4.2 Functional Scope

1. Authentication and role-based access control for officer, risk-analyst, compliance and admin roles.
2. Structured applicant data intake capturing personal, employment, income, debt and loan-request details.
3. Data validation: completeness, format and range checks, with missing-field prompts before a prediction is attempted.

4. Eligibility prediction: a served ML model returning a calibrated probability and risk tier.
5. Explainability: a model-derived, plain-language ranked list of factors behind each prediction.
6. Officer dashboard to view the application, prediction and explanation, and to approve, decline or escalate with a mandatory note on override.
7. Immutable audit log recording every prediction (inputs, model version, output, explanation) and every officer action.

4.3 Data Scope

S. No.	Data / Information	Description
1.	Personal & demographic data	Age, address, number of dependents
2.	Employment & income data	Employment type, tenure and declared income
3.	Existing debt obligations	EMIs and credit-card outstanding, used to derive a debt-to-income ratio
4.	Credit bureau data	Bureau score and summarized report fields such as delinquency history and credit age
5.	Loan request details	Requested amount, tenor and purpose of the loan

4.4 Out of Scope

1. Actual loan disbursal, servicing or collections.
2. Full core-banking or loan-origination-system (LOS) replacement.
3. Fully autonomous, human-free final decisions of any kind.
4. Document OCR/intelligence, fraud-graph analysis and multi-lender marketplace functionality.

Note: Features outside the scope may be mentioned as future enhancements.

5. Technical Feasibility Analysis

Technical feasibility determines whether the proposed system can be technically developed and implemented with the available resources.

5.1 Proposed Technology Stack

Component	Proposed Technology
Frontend	React with JavaScript, Tailwind CSS
Backend	Java (SpringBoot)
Database	PostgreSQL
Programming Language(s)	Python, JavaScript, Java
Development Tools/IDE	VS Code, Git, Docker, GitHub Actions

Component	Proposed Technology
Other Technologies	XGBoost / LightGBM, SHAP (explainability), MLflow (model registry)

5.2 Hardware Requirements

Hardware	Requirement
Processor	Standard multi-core processor (2–4 vCPU sufficient for a pilot deployment)
RAM	4–8 GB for pilot deployment; 8 GB+ recommended for development
Storage	20–50 GB SSD
Other Hardware	No GPU required — gradient-boosted tree models train and serve efficiently on CPU

5.3 Software Requirements

Software	Version / Requirement
Operating System	Linux (containers) for deployment; Windows/macOS/Linux for development
IDE / Editor	VS Code or PyCharm
Database	PostgreSQL 15+
Framework / Libraries	SpringBoot, React, scikit-learn, XGBoost/LightGBM, SHAP, MLflow
Other Software	Docker, Docker Compose, Git

5.4 Technical Skills Required

- Backend/API development (Java (SpringBoot)
- Frontend development (React, JavaScript)
- Applied machine learning (tabular modelling, evaluation, SHAP-based explainability)
- Cloud/DevOps (containerization, CI/CD, monitoring)

Technical Feasibility Assessment

Explain whether the required technology, hardware, software and technical skills are available to the team.

Every component in the proposed stack is mature, widely used in production, and requires no novel research. The technology choices are deliberately conservative: a well-tuned gradient-boosting model on CPU, a relational database for transactionally sensitive records, and open-source, self-hostable tooling (MLflow) to avoid vendor lock-in. No specialized hardware such as a GPU is required. Given

standard development machines and freely available open-source tooling, the project is comfortably within reach of a small, skilled team.

Conclusion

☑Technically Feasible ☐Technically Not Feasible

Justification: The recommended stack (React/TypeScript, FastAPI, PostgreSQL, XGBoost/LightGBM with SHAP, MLflow) is standard, well-documented, and does not depend on scarce or expensive resources.

6. Operational Feasibility Analysis

Operational feasibility determines whether the proposed system can be effectively used and integrated into the intended environment.

6.1 User Acceptance

Officers are likely to accept the system because it is designed around their existing review-and-decide workflow rather than a new paradigm, and because every prediction comes with a plain-language explanation they can inspect rather than a bare score they must simply trust.

6.2 Ease of Use

The officer dashboard surfaces the application, the prediction, and a ranked explanation together on one screen, and requires only a short note when an officer chooses to override the system's suggestion, keeping the added workflow overhead minimal.

6.3 Existing Workflow

The system fits into, rather than replaces, the existing intake → bureau-pull → review → decision workflow: it automates the scoring and explanation step and augments (not replaces) the officer's final decision, which keeps human judgment in the loop for every case.

6.4 Training Requirements

Loan officers would need brief training — on the order of a half-day to a full day — on reading a probability-plus-reason-code output and on when and how to override it. Administrators would need onboarding to user/role management and model-version promotion.

6.5 Expected Operational Benefits

1. Reduced average processing time for high-confidence “clear approve” and “clear decline” cases, freeing officer time for genuinely borderline applications.
2. More consistent reason-giving across officers and branches, since reason codes are model-derived rather than officer-improvised.
3. A complete, reconstructable audit trail for every decision, supporting compliance and dispute resolution.

4. Improved applicant experience through plain-language guidance rather than an unexplained rejection.

Operational Feasibility Assessment

☑Operationally Feasible ☐Operationally Not Feasible

Justification: The system is designed to slot into the existing officer workflow with human review retained at every step, minimizing resistance and change-management risk.

7. Overall Feasibility Summary

Feasibility Area	Status	Justification
Technical Feasibility	Feasible	Conventional, mature and well-supported stack; no novel research required
Operational Feasibility	Feasible	Fits the existing officer workflow with human-in-the-loop review retained

Overall Conclusion

Based on the technical and operational feasibility analysis, the proposed project is feasible because it relies entirely on mature, well-documented technology, fits naturally into the loan officer's existing review workflow, and requires no specialized hardware or unavailable skills. The main considerations for a real-world deployment — representative training data and a jurisdiction-specific compliance review — are business and regulatory conditions rather than technical or operational blockers, and do not prevent the system from being built and demonstrated as proposed.

8. Expected Outcome

The completed software system is expected to deliver a working web application in which a loan officer can submit or review a structured application, receive a calibrated eligibility probability with a ranked, plain-language explanation, confirm or override the recommendation, and rely on an immutable audit log of every prediction and decision. The system should give applicants clearer and faster insight into their eligibility, give officers a defensible basis for every decision they make, and give the wider organization a demonstrable, auditable trail of how each outcome was reached.

9. Future Enhancements (Optional)

1. Interactive “what-if” analysis, letting an applicant or officer adjust an input and see the prediction update live.
2. Loan/product recommendation, suggesting a different amount or product the applicant is more likely to qualify for.
3. Integration of alternative data (with consent) to extend assessment to applicants with thin or no formal credit history.